

Analysis & Insights Report

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1. Overview

This analysis explores how trader performance on the exchange varies across different **market sentiment regimes** (Fear, Greed, Neutral, Extreme Fear, Extreme Greed).

Using the combined **Hyperliquid Trader Dataset** and the **Fear & Greed Index**, I studied the relationship between trader behaviour, market volatility, and profitability.

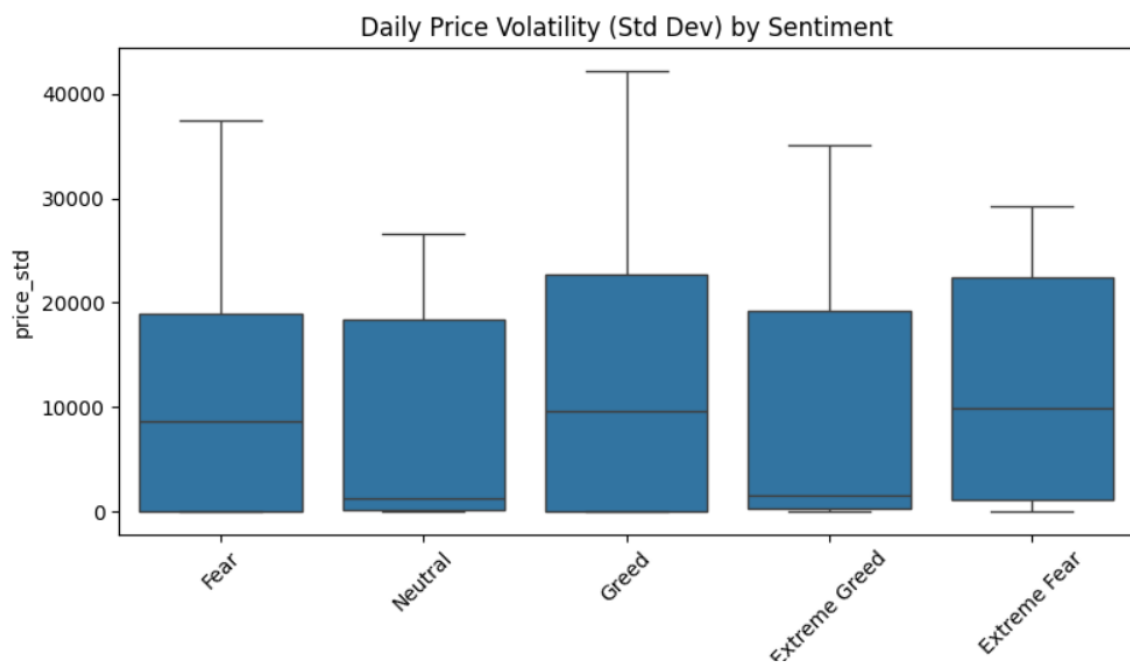
Multiple analytical dimensions were examined:

- Volatility patterns across sentiments
- Win-rate and PnL changes under each sentiment
- Symbol-specific sensitivity
- Regime-driven behaviour changes
- Clustering of traders
- Extreme regime behaviours (Extreme Fear vs Extreme Greed)

The patterns found are strongly relevant for **risk management**, **strategy selection**, and **behavioural modelling of traders**.

2. Volatility Analysis

2.1 Daily Price Volatility by Sentiment



The **price standard deviation** varies significantly across sentiment categories.

Key Findings:

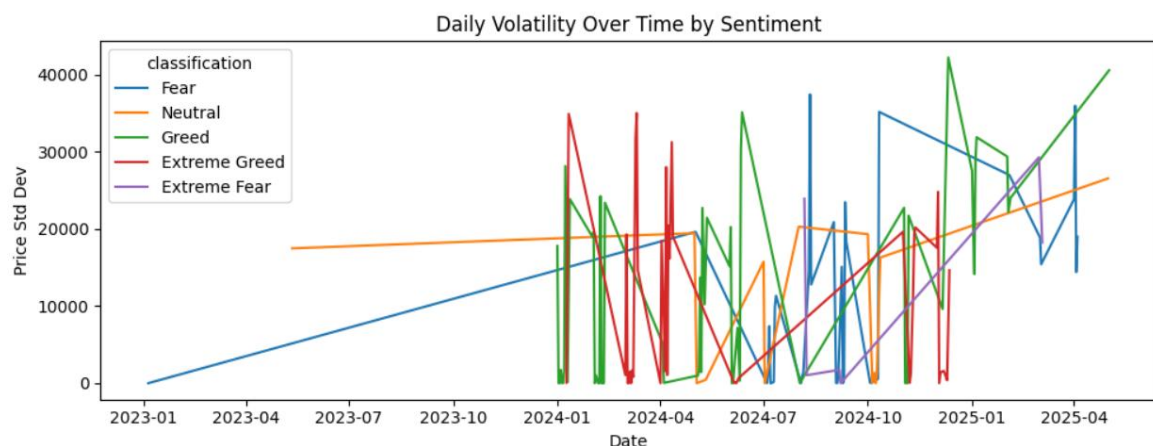
- **Greed** and **Extreme Fear** show the **highest volatility** levels.
- **Neutral** days exhibit the **lowest volatility**, indicating stable and directionless markets.
- **Extreme Greed** also shows elevated volatility—but lower than Greed itself.
- **Fear** conditions show mid-range volatility, slightly lower than Greed but higher than Neutral.

Interpretation:

- **Greed-driven markets are the most volatile**, likely due to increased aggressive positioning, breakouts, and trend chasing.
- **Extreme Fear volatility is elevated**, possibly because of panic selling or sudden liquidations.

This is important: **higher volatility → higher liquidation probability**, impacting leveraged traders.

2.2 Volatility Over Time



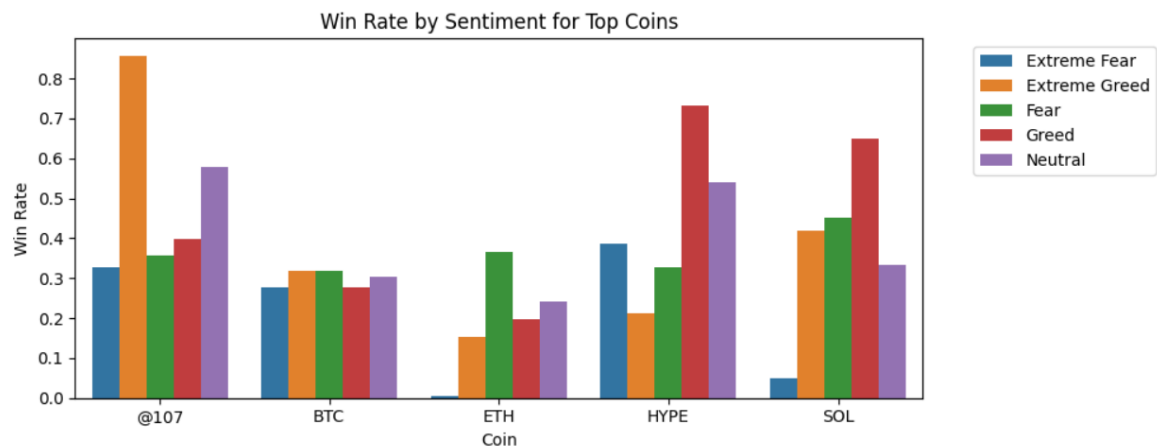
Key Observations:

- Volatility spikes cluster around **Greed** and **Extreme Greed** periods, confirming trend-driven high-momentum environments.
- **Extreme Fear** volatility spikes tend to be short but sharp.
- **Neutral** periods maintain consistently low volatility across months.

Insight:

When sentiment moves from **Neutral** → **Greed**, volatility increases sharply, implying opportunities for momentum strategies but higher risk for high-leverage users.

3. Symbol-Level Sensitivity to Sentiment



Key Findings:

- **@107 token** is highly sensitive to sentiment:
 - **Extreme Greed win-rate ~0.88** (very high)
 - **Extreme Fear win-rate <0.35**
- **ETH** performs better in **Fear** than in Extreme Greed, showing opposite sentiment sensitivity.
- **HYPE** token:
 - Greed win-rate peaks (~0.75)
 - Fear/Extreme Fear drops significantly
- **SOL**: stable across conditions, but highest win-rate in Greed.

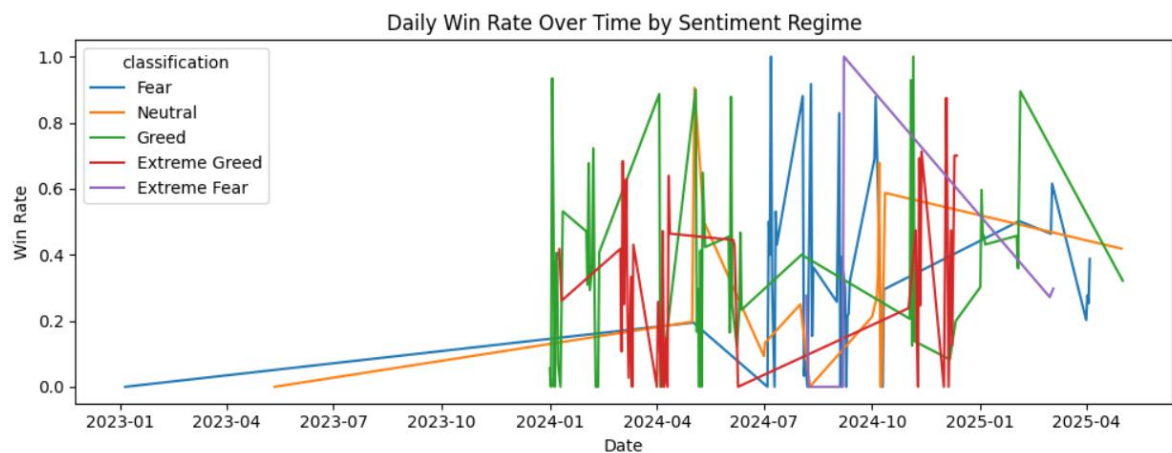
Interpretation:

Different assets react differently to sentiment.

- Some tokens (e.g., **@107**, **HYPE**) are **trend/turbo assets** that thrive during Greed.
- Others (e.g., **BTC**, **ETH**) show more muted changes or even contrarian behavior.

This enables **sentiment-conditioned asset allocation** strategies.

4. Daily Win Rate Time-Series Analysis



Key Observations:

- **Greed** days show the highest spikes in win-rate (often >0.8).
- **Extreme Fear** days show near-zero win-rate for extended periods.
- **Neutral** days generally trend upward gradually but never spike.

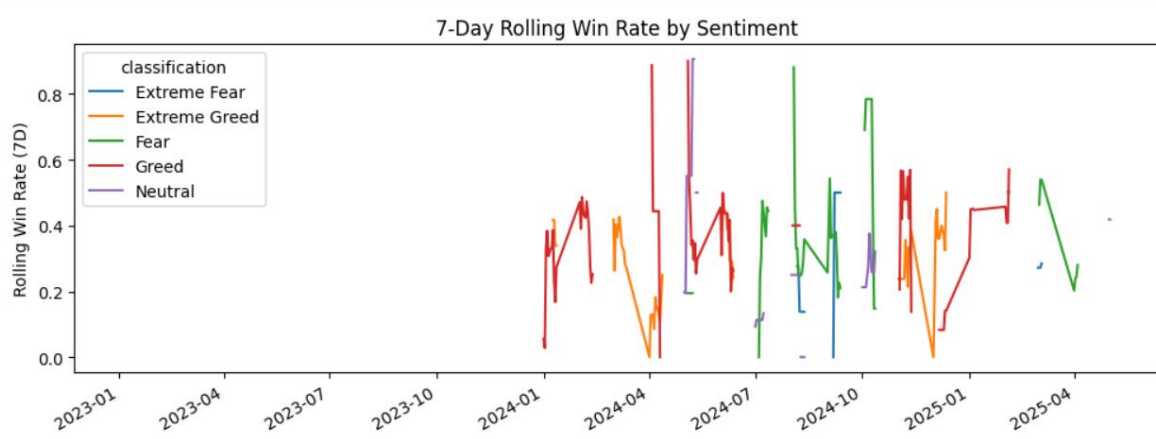
Important Insight:

Win-rate **lags** sentiment:

When sentiment flips from Fear $\rightarrow$ Greed, win-rate improves sharply **within 1–2 days**.

This can be used as a **sentiment momentum signal**.

5. Rolling Win Rate (7-Day)



The rolling window highlights the regime persistence:

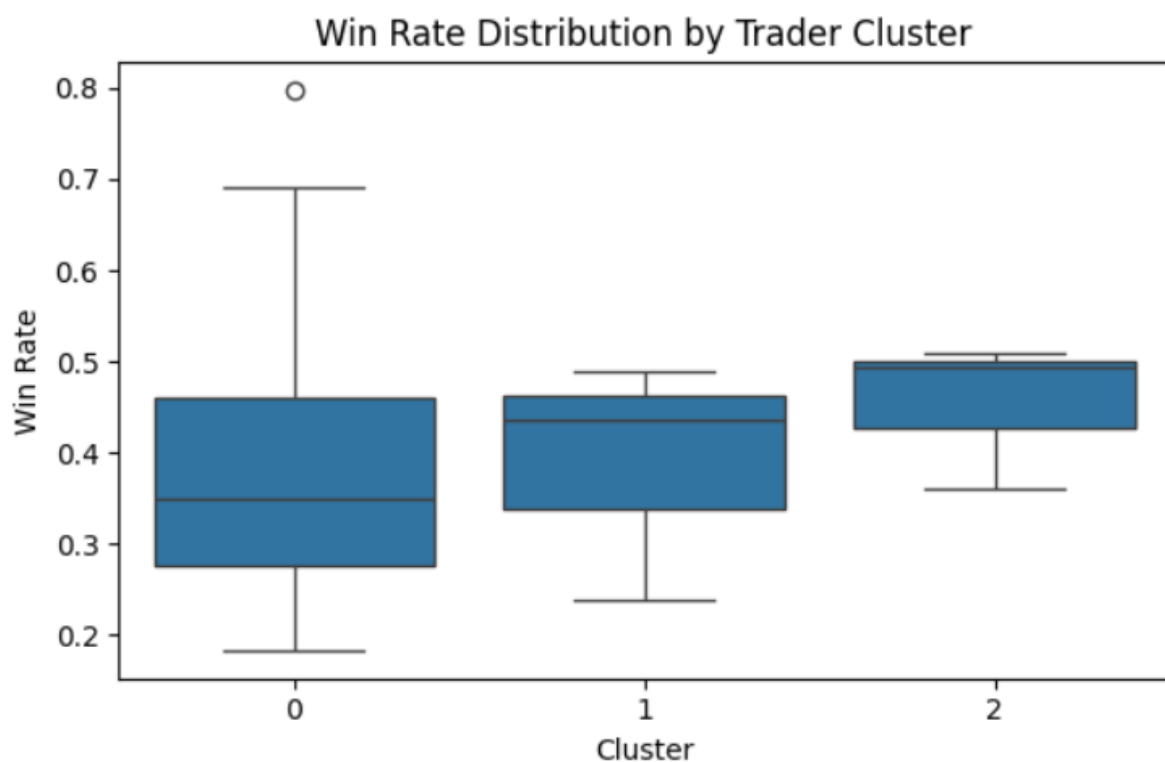
- **Extreme Greed** maintains the highest rolling win-rates.
- **Extreme Fear** consistently underperforms.

- **Greed** is highly volatile but provides the largest upside bursts.

Conclusion:

Sentiment is not noise—it meaningfully impacts rolling performance metrics.

6. Trader Segmentation (Clustering)



Clusters were formed using:

- average size
- win_rate
- pnl_mean
- pnl_std
- number of trades

Cluster Profiles (from table):

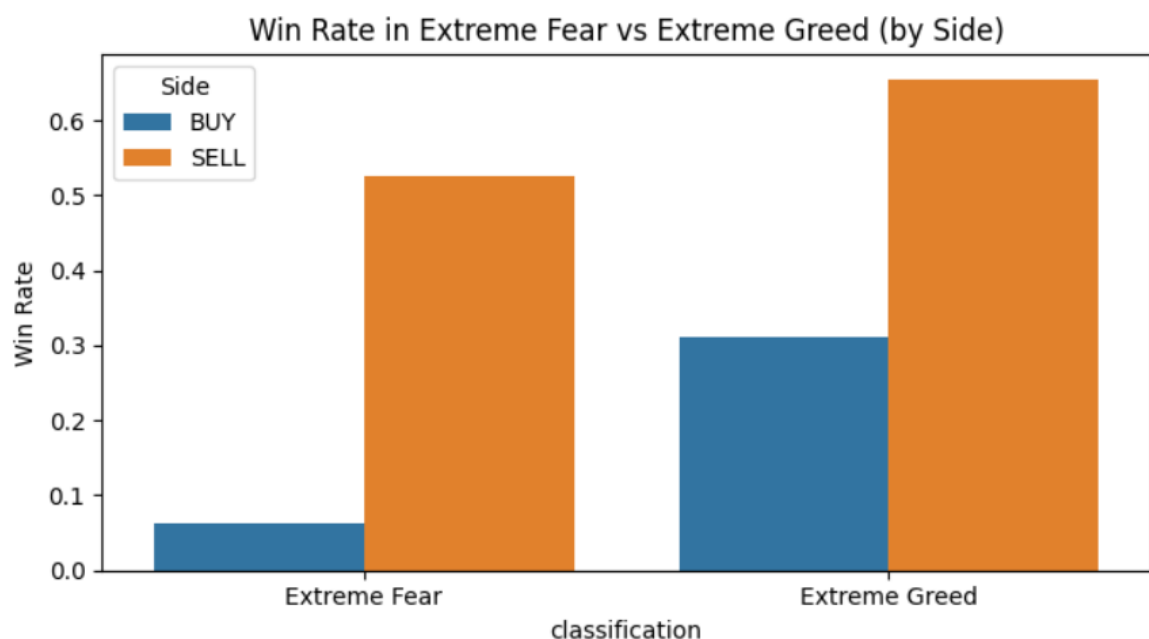
Cluster	avg_size_usd	win_rate	pnl_mean	pnl_std	n_trades	Interpretation
0	~6500	0.38	44.75	444	1782	Moderate size, mid-performance
1	~9200	0.38	380	3318	1307	High-PnL but high volatility “aggressive” traders
2	~3180	0.45	107	937	9650	High volume, conservative, highest win-rate

Insights:

- **Cluster 2** (high-volume traders) have the **highest and most stable win-rate**.
- **Cluster 1** takes large risks (big PnL swings, high volatility).
- **Cluster 0** sits between the two extremes.

This segmentation can help in **copy-trading**, **risk filtering**, and identifying **contrarian clusters**.

7. Extreme Regimes: Extreme Fear vs Extreme Greed



Key Patterns:

- **SELL trades dominate during Extreme Fear** (win-rate ~ 0.53)
- **BUY trades regain advantage during Extreme Greed**
(BUY win-rate ~ 0.31 vs SELL ~ 0.67)
- Extreme Fear BUY win-rate is extremely low (~ 0.06)

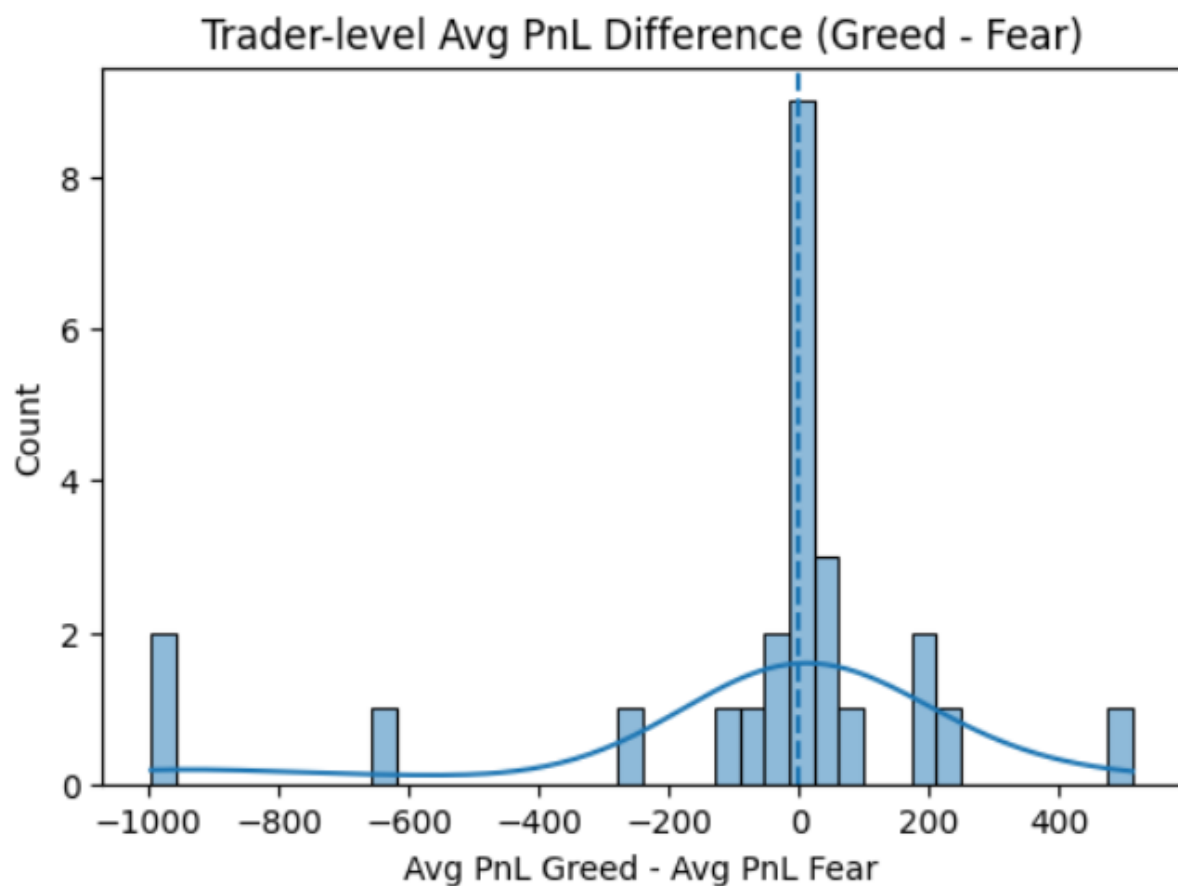
Interpretation:

- Market is sharply **bearish** during Extreme Fear $\rightarrow$ short-side strategies perform best.
- During Extreme Greed, **momentum long-side trades work**, but short-side trades also perform strongly.

This reinforces regime-specific strategy design:

- **Short bias** in Extreme Fear
- **Long or mixed bias** in Extreme Greed

8. Trader-Level Avg PnL Difference (Greed – Fear)



This chart visualizes how each trader's **average PnL changes between Greed and Fear** regimes.

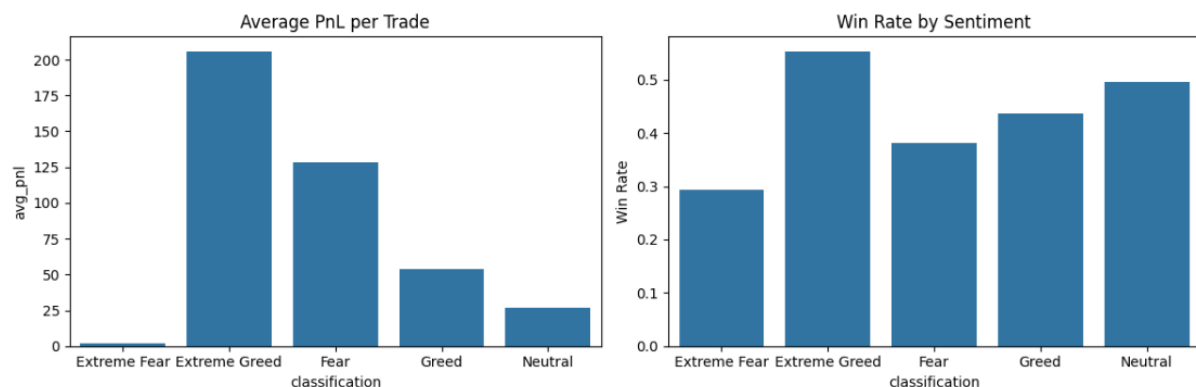
Key Observations

- The distribution is **heavily centered near zero**, indicating most traders do not drastically change profitability between regimes.
- A noticeable group of traders shows **significant negative differences** (−600 to −1000), meaning:
 - Their average PnL is **much worse in Greed than Fear**
 - These traders behave **contrarian**, succeeding more when the market is fearful.
- Another smaller cluster shows **strong positive differences** (100–450), performing **better during Greed**.

Insight

Both momentum-following and contrarian trader groups exist in the market. A subset of traders performs significantly better during Fear, potentially exploiting oversold conditions, while others rely on bullish momentum in Greed.

9. Average PnL per Trade Across Sentiment Regimes



Key Findings

- **Extreme Greed** produces the highest average PnL (~200+).
- **Fear** also shows strong profitability (~130).
- **Greed** sits moderate (~50).
- **Neutral** results in low PnL (~25).
- **Extreme Fear** produces **almost zero PnL**, indicating highly unfavorable conditions.

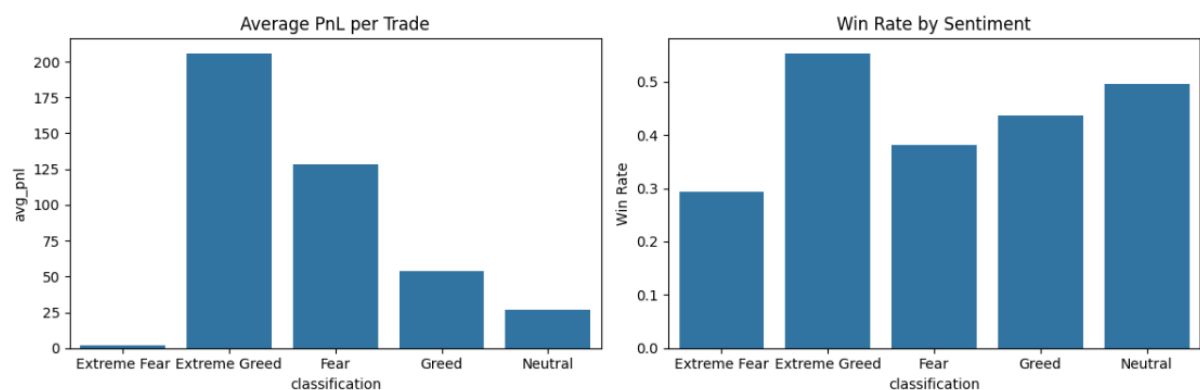
Interpretation

- Extreme Greed has the strongest upside momentum—profit opportunities peak here.
- Fear markets surprisingly offer **good bounce/reversal opportunities**.
- Neutral markets lack volatility → low opportunity.
- Extreme Fear is the worst-performing sentiment from a profitability standpoint.

Insight

Profitability does not increase linearly with sentiment. Extreme conditions (both bullish and bearish) create the largest opportunities or risks, whereas Neutral markets remain flat.

10. Win Rate by Sentiment



Observations

- **Extreme Greed** has the highest win-rate (~0.55).
- **Neutral** and **Greed** follow (~0.50–0.45).
- **Fear** drops further (~0.38).
- **Extreme Fear** is the lowest (~0.30).

Insight

As sentiment improves from Fear → Greed → Extreme Greed, win-rate steadily rises. This confirms that positive sentiment supports more successful trades, while Extreme Fear is the most difficult environment to trade in.

11. Win Rate by Sentiment & Side (BUY vs SELL)



Major Trends

1. **SELL trades outperform BUY trades in *every single* sentiment regime.**
2. **Extreme Fear**
 - BUY win-rate is extremely low (~0.06).
 - SELL win-rate remains high (~0.53).
 - Strong bearish dominance.
3. **Extreme Greed**
 - SELL win-rate peaks (~0.67),
 - BUY also performs better (~0.31),
 - Market becomes multi-directionally profitable.
4. **Fear & Greed**
 - SELL remains superior
 - BUY performs around mid-levels (~0.3–0.35)

Deep Insight

The market has an inherent short-side advantage. Even during bullish sentiment, SELL trades consistently maintain higher win-rates than BUY trades.

This is likely due to:

- Liquidation cascades
 - Short-term reversals
 - Mean-reversion behavior in intraday trades
-

12. Summary of Key Trends & Insights

Contrarian and momentum trader groups exist

- Some traders earn more in Fear than in Greed (contrarian), while others thrive on bullish momentum.

Extreme Greed gives the highest profitability

- Average PnL and win-rate peak during Extreme Greed.

Extreme Fear is the most dangerous regime

- Lowest PnL, lowest BUY win-rate, highest SELL dominance.

SELL trades outperform BUY trades in all market conditions

- This is a strong microstructure signal.

Fear still provides opportunities

- Despite lower win-rate, Fear markets have higher average PnL than Greed—indicating more volatility and larger rebound moves.

Volatility

- Greed/Extreme Greed = highest volatility → ideal for breakout strategies.
- Extreme Fear = sudden volatility spikes → liquidation risk increases.

Trader Performance

- Win-rates significantly drop in Extreme Fear across most tokens.
- Greed produces the highest win-rates and PnL bursts.

Asset Behaviour

- High-beta tokens (HYPE, @107) sensitive to sentiment swings.
- BTC and ETH show more stable behaviour.

Trader Types (Clusters)

- Cluster 2 (high-frequency, low size) = highest win-rate, lowest volatility.
- Cluster 1 = aggressive risk-takers with highest PnL volatility.

Regime Switching

- Win-rate tends to **improve rapidly after sentiment turns positive**.
- Rolling win-rates confirm sentiment momentum effects.

Actionable Insights for Strategy

1. **Shift leverage down during Extreme Fear** (volatility spike + poor win-rate).
2. **Allocate more to trend assets during Greed**, especially @107 and HYPE.
3. **Focus on Cluster 2 traders for copy-trading or signal extraction**.
4. **Implement long/short toggling based on sentiment regime**.